SEQUENCE 5

Term review and remediation

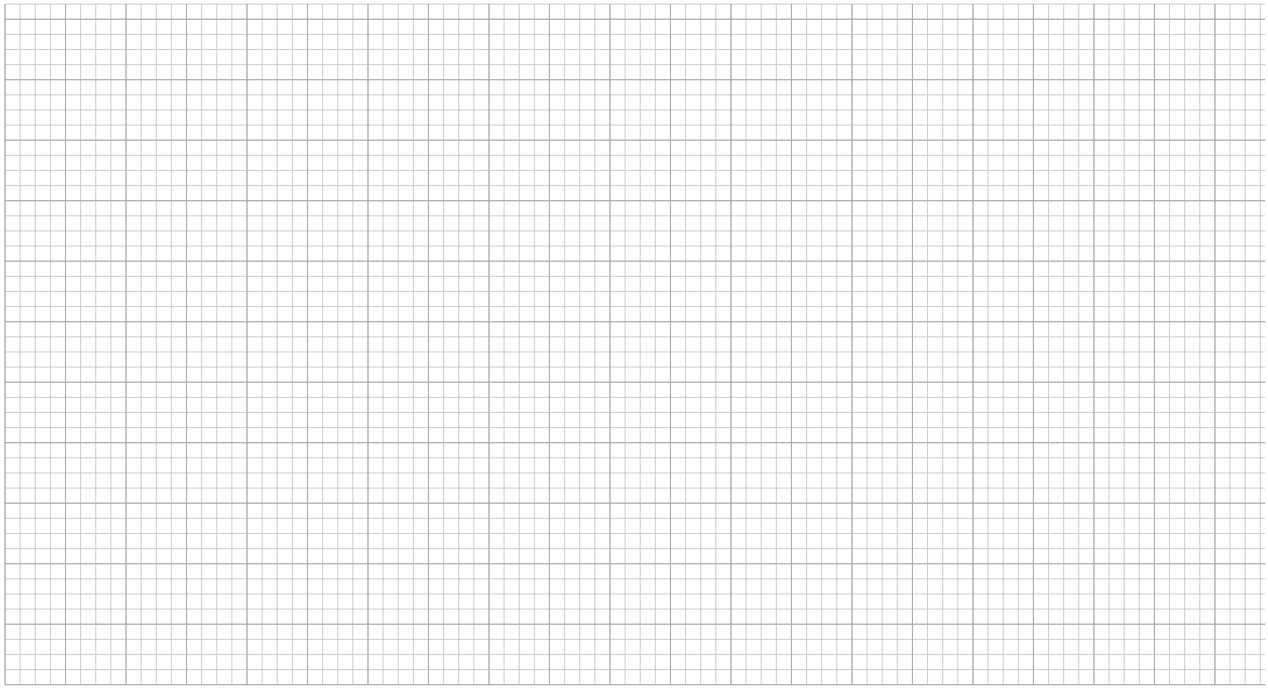
16–20 November 2026

5.1 Individual review

Method · Instructions. Work without the course notes for 25 minutes, then check with the methods in the booklet. For each question, note: solved alone, solved with help, or to redo. Justify the answers.

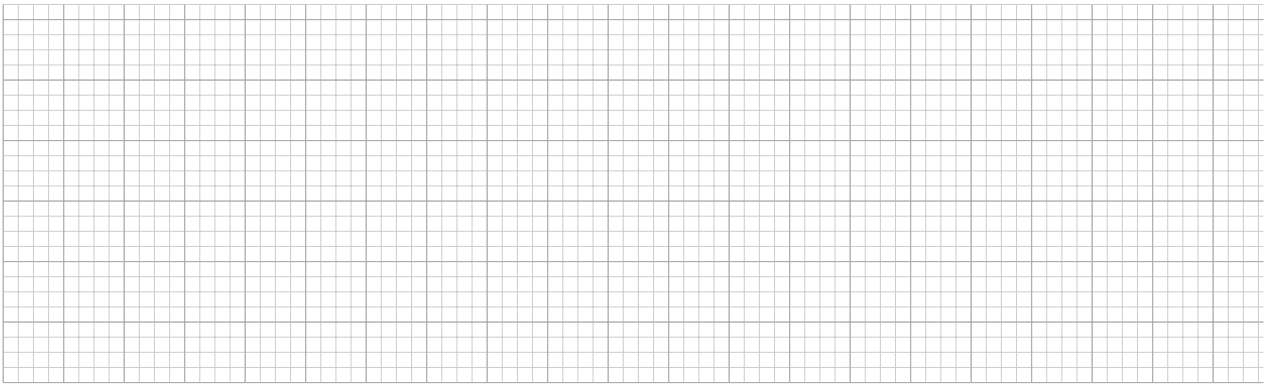
Example 1 · Review of sequences 1 to 4.

1. Reduce $90/126$ to lowest terms. Prove that the sum of two odd integers is even.
2. With $A(-1,2)$ and $B(3,-2)$, calculate $\overrightarrow{AB}$, AB and the midpoint of $[AB]$ in an orthonormal frame.
3. Determine an equation of (AB); does $C(5,-4)$ belong to this line?
4. Calculate $[-2,4]\cap(1,7]$ and $[-2,4]\cup(1,7]$.
5. Solve $|x-1|\leq 4$ and explain why $1/3\notin\mathbb{D}$.



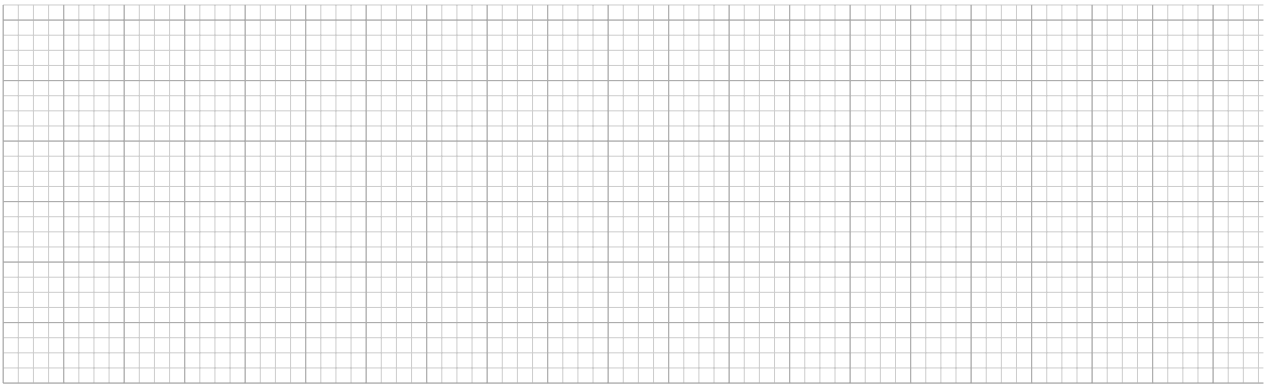
5.2 Correcting an error and revisiting a method

Example 2 · Analysis of student work. Correct: (a) "$\overrightarrow{AB} = (x_A - x_B, y_A - y_B)$"; (b) "The zero vector directs every line"; (c) "$|x - 2| \leq 3$ gives $x \in [2, 3]$"; (d) "$1/3 \approx 0.333$, so $1/3$ is decimal".



Example 3 · Remediation track.

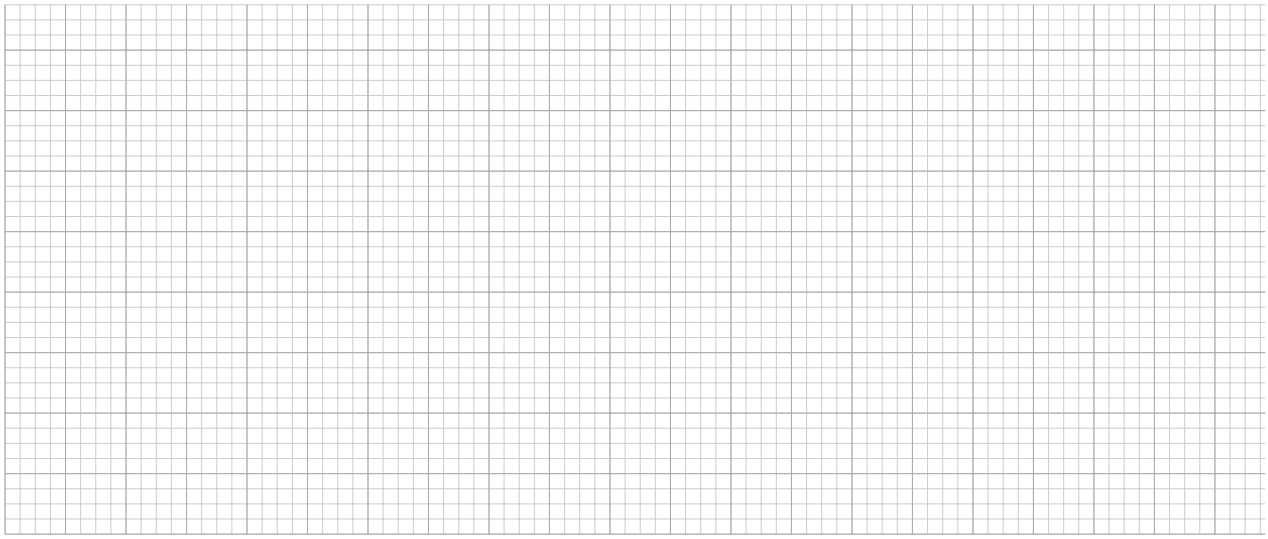
1. Reduce $72/108$, then $\frac{3}{4} - \frac{1}{6}$.
2. With $P(1, 1)$ and $Q(5, 3)$, determine the midpoint, a direction vector and an equation of (PQ).
3. Determine $(-\infty, 3] \cap [-1, +\infty)$, then solve $|x + 2| < 1$.



5.3 Applying and self-assessing

Example 4 · Synthesis problem. Two displacements are represented by $\vec{u}(6, 4)$ and $\vec{v}(9, 6)$.

1. Show that they are collinear and find the coefficient relating $\vec{v}$ to $\vec{u}$.
2. The starting point is $A(-2, 1)$. Determine the endpoints B and C associated with the displacements $\vec{u}$ and $\vec{v}$.
3. Find an equation of the line through these three points.
4. A point $M(x, y)$ of this line must satisfy $|x - 4| \leq 3$. Give the interval of possible abscissae.



Key point · My review. My priority: .....

The mistake to avoid: .....

The method to reuse: .....

An exercise to redo without help: .....